HiSET Mathematics strategy guide

Part 1 works for any test. Part 2 turns those habits into math-specific moves.

> **How to use this guide.** Read Part 1 once, then keep it nearby while you take practice tests. In Part 2, find the content area you miss most often and practice its strategies first. Each area has a worked example you can try before you read the solution.

Part 1. Strategies for any test

Before test day

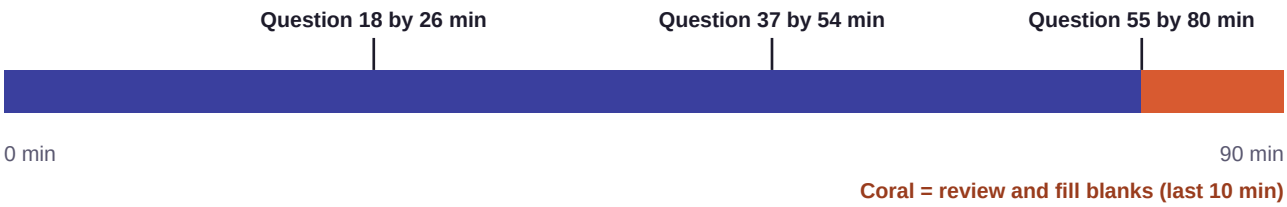
- **Know the format.** How many questions, how much time, whether you can skip and come back, and what tools are allowed.
- **Take at least one timed practice test.** The clock feels different when you have practiced with it.
- **Rest beats cramming.** Do a light review the night before and get a full night of sleep.
- **Plan the logistics.** Bring the ID your test center requires, know the route, and arrive early.
- **Eat and drink something.** A hungry brain loses focus fast.

During the test: six habits

Habit	What it looks like
1. Read the whole question	Underline what is being asked. Circle words like **NOT**, **EXCEPT**, **LEAST**, and **greatest**.
2. Easy ones first	Answer what you know right away. Mark the hard ones, move on, and come back. You can move back and forth between questions.
3. Eliminate	Cross out answers that are clearly wrong. Removing even one raises your odds if you have to guess.
4. Never leave a blank	Your score is based on the number of questions you answer correctly, so a guess costs you nothing.
5. Watch the clock	Use checkpoints (see below) instead of checking every minute.
6. Check before you move on	Reread the question. Did you answer what was asked? Does the answer make sense?

Pacing: a checkpoint plan for a 90-minute, 55-question test

That is about 98 seconds per question if you use every minute. Aim for about 87 seconds so you keep 10 minutes for review.



When nerves show up

- **Box breathing:** in for 4 counts, hold for 4, out for 4. Repeat three times.
- **Reset your body:** drop your shoulders, unclench your jaw, and shake out your hands.
- **Talk to yourself like a coach:** say "I can do the next step," not "I can't do this."
- **Stuck?** Mark it, skip it, and give your brain time to work in the background.

Learn from every practice test

Do not just check your score. Sort each missed question into a category, then fix the category.

Type of mistake	Looks like	Fix
Careless	You knew it but made a sign or arithmetic slip.	Slow down on the last step. Always do a quick check.
Misread	You solved a different question than the one asked.	Underline the question and reread it before you answer.
Concept gap	You did not know how to start.	Study that topic, then do 5 similar problems.
Time	You rushed or ran out of time.	Practice with the checkpoints. Skip and return sooner.

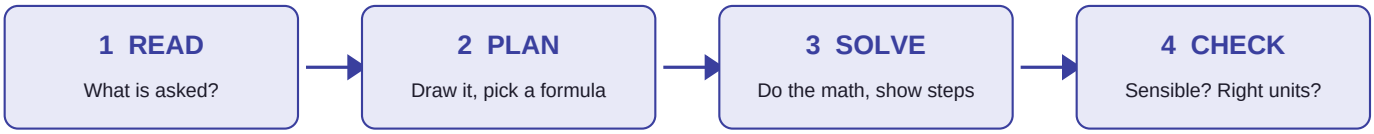
Part 2. Strategies for HiSET Mathematics

The test at a glance

Feature	Details
Questions and time	55 multiple-choice questions in 90 minutes
Content categories	Numbers and operations; measurement and geometry; data analysis, probability, and statistics; algebraic concepts
Calculator	"Calculator neutral." You do not need one, but one is available at the test center. The type depends on the center and format.
Formula sheet	A reference sheet of formulas is provided, but it will not tell you which formula to use.
Passing	A scaled score of at least 8 (out of 20) on each subtest, and at least 45 total across all five.

Details can change. Confirm them in the current HiSET *Test at a Glance* or with your test center.

The four-step problem-solving method



- **Read:** underline the question. Cross out information you do not need.
- **Plan:** draw a picture, write the formula, or set up an equation before you touch the calculator.
- **Solve:** work in steps on scratch paper. Do not do everything in your head.
- **Check:** estimate, plug your answer back in, and confirm the units.

Calculator-neutral strategy

- **Estimate first.** Round the numbers and get a rough answer. If the calculator disagrees by a lot, you pressed something wrong.
- **Use it for the heavy arithmetic,** not for thinking. Set up the problem yourself.
- **Check the display.** Misplaced decimals and missing negative signs are the most common errors.
- **Practice without it too.** Many problems are quicker with friendly numbers (halves, tens, hundreds).

> **Example.** You calculate 48% of 205 and get 984. Estimate: 48% is about half, and half of 200 is 100. 984 is far too big. The correct answer is 98.4. The decimal point slipped.

Turn words into math

Words	Math	Example
is, was, equals, gives	=	"The total is 20" becomes total = 20
sum, total, more than, increased by	+	"7 more than x" becomes x + 7
difference, less than, decreased by	–	"5 less than x" becomes **x** – **5** (order flips!)
product, times, twice, of (with fractions or percents)	×	"25% of 80" becomes 0.25 × 80
quotient, per, out of, split evenly	÷ or fraction	"12 out of 30" becomes 12/30

Two shortcuts when you are stuck

- **Plug in the answer choices (back-solve).** For 3x + 5 = 20, try the choices. x = 5 gives 15 + 5 = 20. Done.
- **Pick easy numbers.** If a problem uses a variable and the question is about patterns, try x = 2 or x = 10 and see which choice works.

Numbers and operations

Covers: integers, fractions, decimals, percents, ratios and proportions, exponents and roots, scientific notation, order of operations, estimation.

Key strategies

- **Convert to one form** (all decimals or all percents) before you compare numbers.
- **Percent = part ÷ whole × 100.** Identify the part and the whole first.
- **Percent change = (new – old) ÷ old × 100.** Always divide by the *original* amount.
- **Ratios:** add the parts to get the total, then find the value of one part.
- **Order of operations:** parentheses, exponents, multiply and divide (left to right), then add and subtract (left to right).

3/10 = 0.3 = 30%

Same amount, three names.
Convert to one form before comparing.

Fraction	Decimal	Percent
1/2	0.5	50%
1/4	0.25	25%
3/4	0.75	75%
1/5	0.2	20%
1/8	0.125	12.5%
1/3	0.333...	33.3%

Watch out for

- Using the *new* amount instead of the original as the base for percent change.

- Ordering negatives: –5 is *less* than –2.
- Adding fractions by adding the denominators. Find a common denominator instead.

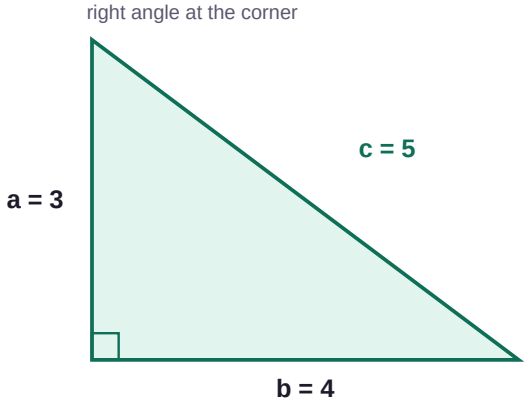
Try it. A jacket that costs $80 is 25% off. What is the sale price?
Solution. 25% of 80 is 20, so the sale price is 80 – 20 = **$60**. Shortcut: you pay 75%, and 0.75 × 80 = 60.

Measurement and geometry

Covers: perimeter, area, volume, circles, angles, triangles, the Pythagorean theorem, similar figures, unit conversion, coordinate geometry.

Key strategies

- **Draw and label** the figure, even if one is given. Add the numbers from the problem.
- **Write the formula first,** then substitute. It prevents most mix-ups.
- **Track units.** Area uses square units and volume uses cubic units.
- **Angle sums:** triangle = 180°, straight line = 180°, quadrilateral = 360°.
- **Similar figures:** set up a proportion with matching sides in the same order.
- **Convert units** by multiplying by a fraction equal to 1, such as 12 in / 1 ft.



Pythagorean theorem

$a^2 + b^2 = c^2$

9 + 16 = 25

c = square root of 25 = 5

c is the longest side, across from the right angle.

Watch out for

- **Radius vs. diameter.** The radius is half the diameter. Circle area uses the radius.
- **Area vs. perimeter.** Perimeter is the distance around. Area is the space inside.
- **Pythagorean theorem:** the hypotenuse is the longest side. Do not just add the legs.
- Mixing units (feet with inches) in the same calculation.

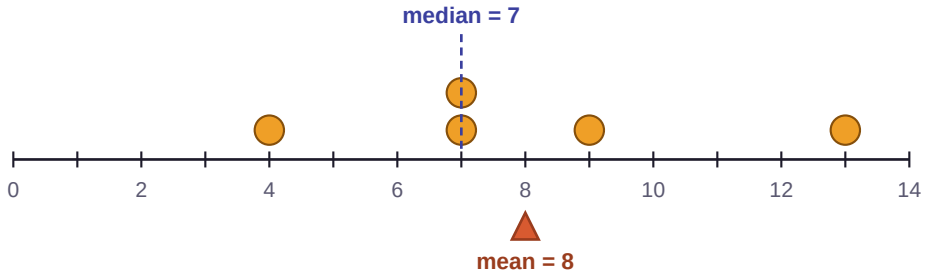
Try it. The legs of a right triangle are 6 and 8. How long is the hypotenuse?
Solution. $6^2 + 8^2 = 36 + 64 = 100$, and the square root of 100 is **10**.

Data analysis, probability, and statistics

Covers: mean, median, mode, range, tables, graphs, scatterplots and trends, probability, counting, simple and compound events.

Key strategies

- **Read the graph before the question:** title, axis labels, units, and the scale (does it start at 0? count by 5s?).
- **Mean** = sum ÷ number of values. **Median** = middle value *after putting the data in order* (average the two middle values if there is an even count).
- **Probability** = favorable outcomes ÷ total outcomes. It is always between 0 and 1.
- **Not happening:** P(not A) = 1 – P(A).
- **Counting principle:** multiply the choices. 3 shirts and 4 pants make 3 × 4 = 12 outfits.



Data: 4, 7, 7, 9, 13

Mode = 7 (appears most)

Range = 13 - 4 = 9

Watch out for

- Finding the median without ordering the numbers first.
- Graphs with scales that jump (by 5s or 10s) or do not start at zero.
- Answers that include the word "probability" but are bigger than 1. Recheck your fraction.
- Whether items are replaced (probability stays the same) or not (the total changes).

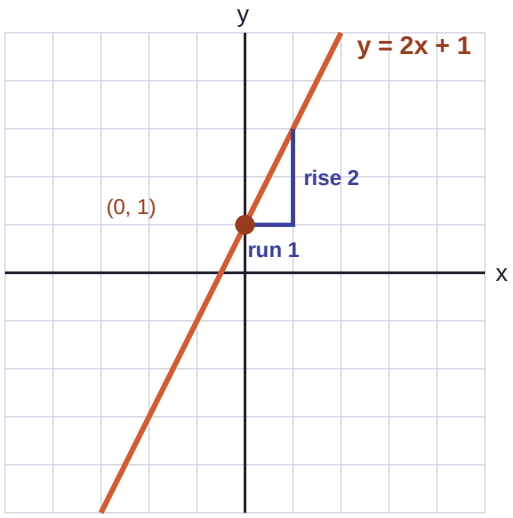
Try it. A bag has 4 red, 6 blue, and 5 green marbles. What is the probability of drawing a marble that is *not* blue?
Solution. There are 15 marbles. Not blue = 4 + 5 = 9, so the probability is 9/15 = **3/5**.

Algebraic concepts

Covers: expressions, linear equations and inequalities, slope and graphing lines, systems of equations, patterns, exponents, factoring and quadratics, functions.

Key strategies

- **Keep it balanced.** Whatever you do to one side of an equation, do to the other.
- **Undo operations in reverse:** deal with adding and subtracting first, then multiplying and dividing.
- **Distribute with care.** A negative outside parentheses changes every sign inside.
- **Flip the inequality sign** when you multiply or divide both sides by a negative number.
- **Slope** = $(y_2 - y_1) \div (x_2 - x_1)$.
- **Systems of equations:** use substitution or elimination, then plug back in to check.



y = mx + b

m = slope = rise / run

b = where the line crosses the y-axis

Here: m = 2 and b = 1

Positive slope: line goes up to the right.
Negative slope: line goes down to the right.

Watch out for

- Sign errors, especially with subtraction and negatives.
- Forgetting to flip the inequality sign.
- Mixing up slope (m) and the y-intercept (b).
- Solving for x when the question asks for something like 2x + 1. Answer what is asked.

Try it. Find the slope of the line through (0, 1) and (2, 5), then write its equation.
Solution. Slope = (5 – 1) ÷ (2 – 0) = 4 ÷ 2 = **2**. The line crosses the y-axis at 1, so **y = 2x + 1**.

Quick reference

Know these cold

The test provides a formula sheet, but you will work faster if you do not need it for the basics.

Topic	Know
Percent	percent = part ÷ whole × 100; percent change = (new – old) ÷ old × 100
Statistics	mean = sum ÷ count; median = middle (ordered); mode = most often; range = highest – lowest
Probability	favorable ÷ total; P(not A) = 1 – P(A)
Lines	slope = rise ÷ run; y = mx + b
Area	rectangle = l × w; triangle = (1/2) × b × h; circle = pi × r^2
Circle	circumference = 2 × pi × r (use 3.14 for pi unless told otherwise)
Volume	box = l × w × h
Triangles	angles add to 180°; $a^2 + b^2 = c^2$ for right triangles
Conversions	12 in = 1 ft; 3 ft = 1 yd; 5,280 ft = 1 mi; 16 oz = 1 lb; 4 qt = 1 gal; 60 min = 1 hr

Your last-10-minutes checklist

- Fill in **every** blank. Guess using elimination.
- Return to the questions you marked, easiest first.
- Recheck questions with **NOT**, **EXCEPT**, or **LEAST** in them.
- On algebra, plug your answer back into the original equation.
- On geometry, check your units and that you used the radius, not the diameter.
- On word problems, reread the last sentence: did you answer *that*?

> **Remember:** you do not need a perfect score. You need 8 out of 20 on each subtest, so steady, accurate work on the questions you can do matters more than racing through the hardest ones.